

Digital Gift Card & Loyalty Point Exchange

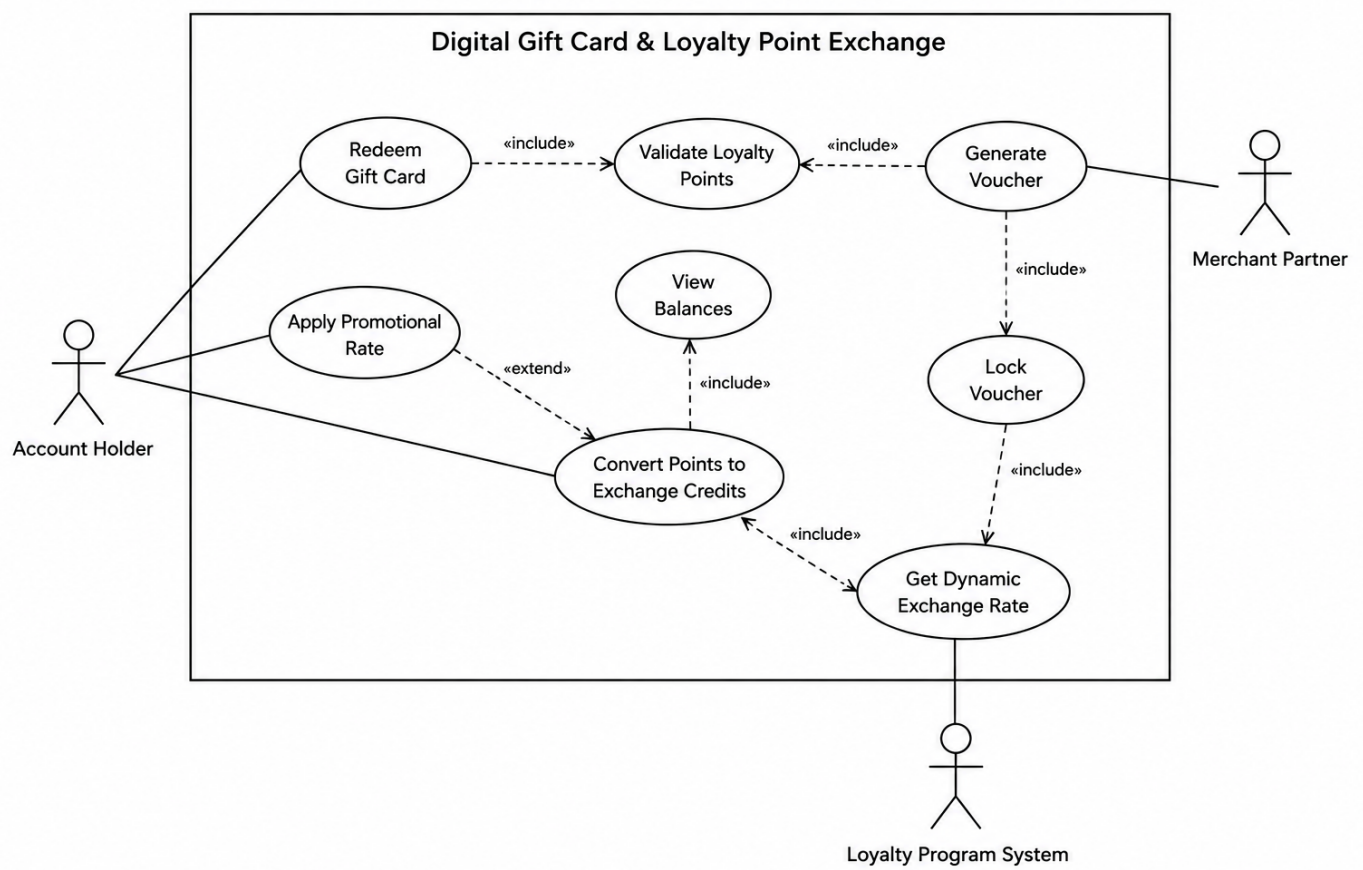
Requirements Engineering & UML Use-Case Modelling — Solution

1. Requirements Table

Exactly 5 functional requirements and 2 non-functional requirements, following the lab requirements.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	High	Pass: valid points produce tokens and a voucher with a cryptographic checksum. Fail: invalid or duplicate data is rejected.	Core conversion and voucher function.
FR-002	Functional	The system shall allow an account holder to view current loyalty-point and universal exchange-credit balances.	Medium	Pass: displayed balances match the ledger after a successful transaction. Fail: stale or inconsistent balance is shown.	Provides visibility of available value.
FR-003	Functional	The system shall apply the current dynamic exchange rate when converting eligible loyalty points into universal exchange credits.	High	Pass: the rate effective at transaction time is used and recorded. Fail: conversion is blocked if no valid rate exists.	Dynamic rates are a core platform function.
FR-004	Functional	The system shall allow an account holder to redeem a valid digital gift-card voucher using available universal exchange credits.	High	Pass: sufficient balance is debited once and voucher becomes redeemed. Fail: insufficient balance or invalid voucher prevents redemption.	Primary end-user value.
FR-005	Functional	The system shall lock a voucher before redemption and reject vouchers that are expired, already locked, or already redeemed.	High	Pass: only one redemption attempt obtains the lock. Fail: conflicting or invalid voucher is rejected.	Prevents duplicate redemption and fraud.
NFR-001	Nonfunctional	Point conversion and gift-card redemption operations must process within 250 ms with zero ledger balance drift.	High	Pass: tests meet ≤ 250 ms and reconciliation shows zero balance drift under simulated peak load.	Ensures performance and financial consistency.
NFR-002	Nonfunctional	The system shall protect voucher codes and transaction records using authenticated access and encrypted data in transit and at rest.	High	Pass: unauthorized access is denied and security tests confirm encryption.	Protects vouchers and transaction data.

2. UML Use-Case Diagram



3. Use-Case Flow Specification — UC-04 Redeem Gift Card

Use Case	UC-04 — Redeem Gift Card
Primary Actor	Account Holder
Supporting Actor	Merchant Partner
Preconditions	Account is authenticated; voucher is valid; sufficient universal exchange credits are available.
Postconditions	Credits are debited exactly once; voucher is marked redeemed; transaction is recorded.

Main Success Scenario

- 1. Account Holder selects **Redeem Gift Card** and enters/scans the voucher code.
- 2. System validates voucher status, expiry, and format.
- 3. System **locks the voucher** to prevent concurrent or duplicate redemption.
- 4. System checks the available universal exchange-credit balance.
- 5. System debits the required credits from the account ledger.
- 6. System marks the voucher as redeemed and records the transaction.
- 7. System displays confirmation and the updated balance.
- 8. Use case ends successfully.

Alternate Flow — Invalid / Already Used Voucher

- 3a. If the voucher is expired, already locked, or already redeemed, the system rejects the attempt.
- 3b. No exchange credits are debited.
- 3c. The system displays an error message and the use case ends without changing the account balance.